

Harshit Rai

Mobile: +91 88401 56900 | Email: raiharshit900@gmail.com | LinkedIn: linkedin.com/in/harshitr10 | GitHub: github.com/Harshitr10

SUMMARY

Backend Software Engineer with 2 years owning distributed, transaction-heavy systems end-to-end in Java, Spring Boot, and Node.js. Independently designed, built, and shipped queue-driven services (RabbitMQ, Kafka) that cut processing latency 20%, reduced manual operational effort 70%, and hit 90% success rates on critical integrations.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, SQL

Backend & Messaging: Spring Boot, Spring Batch, RabbitMQ, Kafka, Redis, Node.js, Express.js, NestJS, REST APIs, Microservices, JUnit, Mockito

Databases: MongoDB, MySQL, PostgreSQL

Cloud & DevOps: AWS (EC2, S3), Linux, Git

AI/ML: Spring AI, LLM Integration, RAG (Retrieval-Augmented Generation), Vector Databases (pgvector)

EXPERIENCE

RapiPay Fintech

Software Engineer

Noida, UP

July 2025 - Present

- **BBPS Bill Payment Integration:** Integrated BBPS (Bharat Bill Payment System) bill-fetch and bill-payment flows across multiple billers and service categories, with an automated **refund service** to reverse failed transactions, achieving a **90%** success rate across billers through resilient retry and error-handling logic on top-up and payment confirmation calls.
- **Reconciliation Automation Service:** Built an automation service that compares the company's ledger, service-wise, against external bank statements to surface extra debited/credited transactions and other mismatches across multiple banking partners, flagging discrepancies for review instead of manual **Excel-based** cross-checking; cut reconciliation effort by nearly **70%**.
- **Internal Fund Transfer Service:** Built an internal fund transfer service for the credit team to move company funds across internal accounts, using **queue-based processing** to prevent duplicate requests; routed individual requests as single transfers and switched to **bulk processing** once the amount crossed a configurable threshold per provider and transaction type, with a **maker-checker** approval workflow.

Software Engineer Intern

August 2024 - July 2025

- **Merchant QR Transaction & Settlement Service:** Built a QR generation scheduler with **MCC-based** merchant onboarding and **KYC** verification flow, plus a QR activation service integrated with **soundbox** activation across POS devices; applied **Redis-backed idempotent** webhook handling for real-time error monitoring and transaction history tracking, with **RabbitMQ**-based queue analysis of success/failure rates driving instant **soundbox** payment-confirmation alerts to merchants.

PROJECTS

RapiPay – Queue-Driven Settlement Engine

Company Project · Team of 2

- **Distributed Transaction Processing Engine:** Designed and built a processing pipeline that divided transaction load across concurrent workers, enforcing **idempotency**, **thread safety**, and per-provider rate limits at intake; priority-based queue routing to downstream channels cut average processing latency by **20%** and reduced upstream rate-limit errors by **60%**.
- **State Machine & Event-driven Processing:** Enforced state transitions through an onHold / active / inactive state machine to safely park in-flight transactions pending upstream checks, publishing state-change events to **Kafka** for downstream consumers while channeling active transactions through a priority-based queue for processing.

AI-Based Fraud Detection Engine

Individual Project

- **Rule-Based Scoring Engine:** Built a configurable, deterministic rule engine scoring transactions in real time across amount, velocity, time-of-day, and channel-novelty checks, classifying risk into LOW/MEDIUM/HIGH bands.
- **RAG-Based Risk Explanation:** Architected a Retrieval-Augmented Generation pipeline using **pgvector** to retrieve semantically similar historical fraud cases, grounding **LLM**-generated risk explanations (via **Spring AI**) in confirmed precedent rather than unguided inference.

EDUCATION

KIET Group of Institutions

B.Tech in Computer Science | CGPA: 7.72

Ghaziabad, UP

2021 - 2025